

Def $V \xrightarrow{T} V$ linear map T from a vspace V to itself is called an operator on V .

Then dim formula says $\dim V = \dim \ker T + \dim \operatorname{im} T$

T injective $\Leftrightarrow \ker T = 0 \Leftrightarrow \dim \ker T = 0 \Leftrightarrow \dim \operatorname{im} T = \dim V$

Revise old table!
(of equivalent conditions)

$\Leftrightarrow \operatorname{im} T = V$

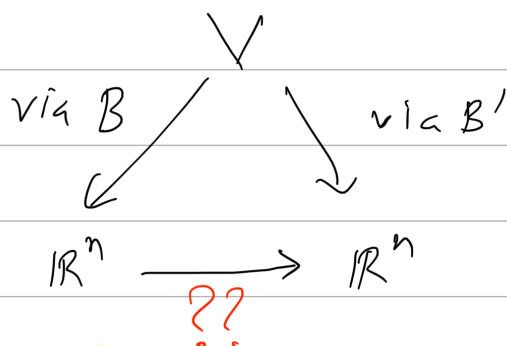
$\Leftrightarrow T$ is surjective.

Change of basis for V .

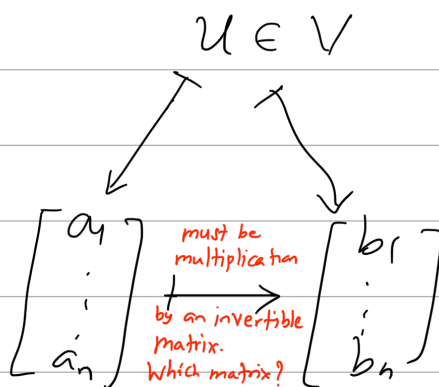
"old" basis $B = (v_1, \dots, v_n)$
 "new" basis $B' = (w_1, \dots, w_n)$ } Change of basis matrix from B to B' is

P = columns are coordinates of new basis vectors with respect to old basis vectors

i.e. $w_1 = p_{11}v_1 + p_{21}v_2 + \dots + p_{n1}v_n$
 $\vdots$
 $w_i = p_{1i}v_1 + \dots + p_{ni}v_n$



Exercise which matrix



i.e. $u = a_1v_1 + \dots + a_nv_n$
 $= b_1w_1 + \dots + b_nw_n$

To find $\begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix}$ from $\begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix}$, substitute for v 's in terms of w 's

Change of basis for a linear map: later! "Answer: $Q^{-1}AP$ "

Section 3-5 Artin

Direct sums (external, internal, how they are "the same" i.e. isomorphic)

Def's

① External direct sum of two vector spaces W_1 & W_2 , denoted $W_1 \oplus W_2$, is a vector space defined as follows

↗ set of ordered pairs

As a set $W_1 \oplus W_2 = W_1 \times W_2$ cartesian product

addition & scalar multiplication: slotwise i.e.

$$(w_1, w_2) + (w'_1, w'_2) := (w_1 + w'_1, w_2 + w'_2)$$

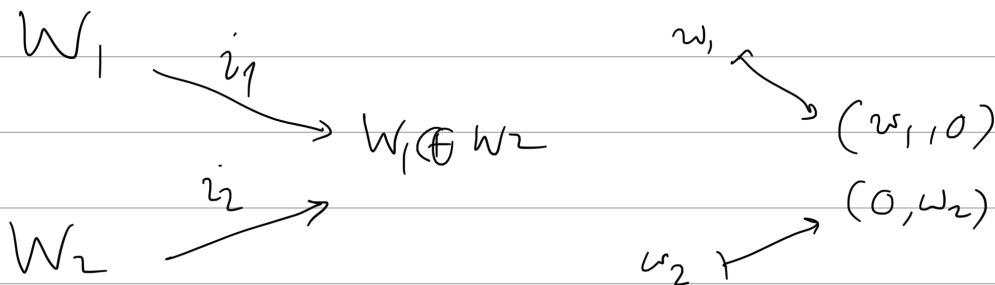
$$c(w_1, w_2) := (cw_1, cw_2)$$

Vector space axioms: straightforward check

e.g. $\mathbb{R}^2 \cong \mathbb{R} \oplus \mathbb{R}$

• We can similarly define $W_1 \oplus \dots \oplus W_k$

• W_1, W_2 are not subspaces of $W_1 \oplus W_2$, but there are injective lin maps



• $\dim(W_1 \oplus W_2) = \dim W_1 + \dim W_2$ Claim

Let basis of $W_1 = (v_1, \dots, v_n)$
 basis of $W_2 = (u_1, \dots, u_k)$
 $\Rightarrow \{(v_i, 0) \mid i=1, \dots, n\} \cup \{(0, u_j) \mid j=1, \dots, k\}$ is a basis for $W_1 \oplus W_2$

Pf. easy

② Defⁿ Suppose $W_1, W_2, \dots, W_k$ are subspaces of V .

$$\begin{aligned} \cdot \text{Span of } (W_1 \cup W_2 \dots \cup W_k) &= "W_1 + W_2 + \dots + W_k" \\ &= \{w_1 + w_2 + \dots + w_k \mid w_i \in W_i \forall i\} \\ &= \text{smallest subspace of } V \text{ containing all } W_i. \end{aligned}$$

• " $W_1, W_2, \dots, W_k$ are lin indep subspaces" means the following: $\left(\begin{array}{l} \text{If } w_1 + w_2 + \dots + w_k = 0 \text{ where } w_i \in W_i \forall i \\ \text{then each } w_i = 0 \end{array} \right)$

Relating $W_1 \oplus W_2$ with $W_1 + W_2$ for W_1, W_2 subspaces of V

We have a linear map $W_1 \oplus W_2 \xrightarrow{S} V$ Image = $W_1 + W_2$
(check linearity: easy) $(w_1, w_2) \mapsto w_1 + w_2$

Apply dimension formula!

$$\dim \text{domain} = \dim \text{image } S + \dim \ker S$$

$\downarrow$

$$\dim W_1 + \dim W_2 = \dim (W_1 + W_2) + ?$$

$$\text{What is } \ker S = \{(w_1, w_2) \mid w_1 + w_2 = 0 \text{ i.e. } w_2 = -w_1\}$$

$$\ker S = \{(w, -w) \mid w \in W_1 \cap W_2\} \quad w_1 \in W_1 \cap W_2 \quad \Downarrow$$

$\downarrow 2 \quad \uparrow$ isomorphism!

$$W_1 \cap W_2 \ni w$$

$$\text{Thus } ? = \dim (W_1 \cap W_2)$$

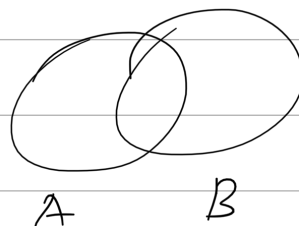
Thus for any subspaces W_1, W_2 in a f.dim v-space V

$$\dim W_1 + \dim W_2 = \dim(W_1 + W_2) + \dim(W_1 \cap W_2)$$

Compare with Artin's direct proof in 3.6.6(a). The above proof is from HW3 problem 1.5 in Chapter 4. So I will replace that problem in HW with another

Analogue to fact about finite sets $A \neq B$

$$|A| + |B| = |A \cup B| + |A \cap B|$$



Exercise

Write formula for $|A \cup B \cup C|$. Analogue for subspaces?